

WORK EXPERIENCE

UGenome.ai , (Remote) - Software Engineer

May 2025 — Present

- Architected and implemented a scalable **FastAPI** microservices backend powering genomics and healthcare analytics, leveraging **asynchronous processing (asyncio, aiohttp)**, **modular routers**, and **dependency injection** for maintainability and high throughput.
- Engineered secure authentication and authorization workflows using **Auth0**, **OAuth2**, and **JWT**, with custom **Pydantic**-based configuration management and centralized **exception handling** to ensure compliance, traceability, and resilience across distributed services.
- Enhanced API reliability and developer productivity by automating **health checks**, **dynamic CORS policies**, **structured logging**, and **environment-driven feature toggles**, supporting rapid iteration and seamless deployment in production environments.
- Implemented scalable **ETL data pipelines** and **asynchronous job management** using **Celery**, **Redis**, and **SQLAlchemy**, with support for **file uploads**, **results tracking**, and **AWS S3** integration for cloud storage and retrieval.
- Utilized **Pandas** and **NumPy** for data preprocessing, statistical computation, and performance optimization within analytical workflows, enabling efficient handling of large genomics and healthcare datasets.
- Developed responsive and user-friendly web interfaces using **React**, **HTML**, and **CSS**, ensuring seamless user experience across devices and browsers.

SenecaGlobal IT Services Pvt Ltd, Hyderabad, India - Technical Analyst

Nov 2023 — Aug 2024

- Engineered scalable **Python-based APIs** using **Django** Framework for the Greenstreet project, enhancing backend architecture and reducing transaction latency by over 5 seconds in high-volume datasets across distributed nodes.
- Optimized complex **SQL queries** and **indexing mechanisms**, boosting database response time by 35% and aligning architecture with modern cloud-native AWS application standards for scalable deployments.
- Integrated **Apache Kafka** into backend data pipelines to enable real-time event streaming, reducing data sync lag across services and supporting faster downstream processing.
- Built dynamic, stakeholder-ready **Tableau dashboards** from fragmented regional data sources, transforming market insights into actionable strategies for improved leadership decision-making and business growth.
- Automated daily **ETL refresh pipelines** using **Apache Airflow** and version-controlled schedules, improving reliability, execution traceability, and reducing data load failures by 60%.
- Implemented **GitLab CI/CD** pipelines to automate build and deployment workflows, enhancing release reliability and reducing manual intervention during software delivery.

IndiaMart Intermesh Ltd , Noida, India - Senior Software Engineer

Jan 2023 — Nov 2023

- Designed and implemented a **Lakehouse architecture** using **Apache Hudi** and **Apache Spark**, delivering significant cost savings and enabling high-speed, scalable analytics for large datasets.
- Developed **scalable RESTful APIs** and backend systems in Python and C++, integrated with Google Cloud Storage and BigQuery to support real-time reporting.
- Optimized modular **Python microservices** and backend architecture, reducing service latency by 40%, improving throughput, and enhancing performance under high-concurrency user traffic.
- Implemented robust **CI/CD pipelines** using **Docker** and **GitLab CI/CD**, accelerating deployment frequency and improving build stability, traceability, and rollback efficiency.
- Engineered a centralized data storage solution that **reduced data silos**, enhanced fraud detection analysis, and supported scalable ML model training with higher data reliability.

Linkedin, Bangalore, India - Software Engineer

Sep 2021 — Jan 2023

- Automated large-scale **job data scraping** using **Python** and **AWS S3**, improving refresh reliability and accelerating weekly labor market insight generation by 5x across key regions.
- Built fault-tolerant, modular **Python scripts** to streamline and schedule data extraction, significantly reducing refresh times and eliminating manual update errors across multiple platforms.
- Modularized and refactored existing **ETL workflows** using **Python**, improving code readability, reducing maintenance overhead, and enhancing long-term pipeline stability in production systems.
- Maintained and enhanced **Git-based CI/CD systems**, increasing collaborative efficiency, deployment consistency, and enabling testable, traceable releases for internal engineering stakeholders.

RWS Moravia (Webdunia), Indore, India- Software Engineer

Jun 2018 — Sep 2021

- Developed scalable APIs and **system-level applications in C++ and Python** to retrieve and process **KPIs** using **Pandas** and **Numpy** for 500+ users,
- Engineered a high-accuracy (85%) text classification model using **Scikit-learn** and **Tensorflow**, significantly reducing data labeling latency and improving real-time content tagging.
- Built a conversational **chatbot** with **IBM Watson** and **Python**, delivering automated diagnostics, reducing manual support by 40%, and enhancing customer experience consistency.
- Containerized production environments using **Docker** and **Travis CI**, enabling seamless deployments, reducing environment-specific errors, and improving build stability under load.

EDUCATION

University of Arizona - Master's, Information Science (Machine Learning)

Rajiv Gandhi Proudlyogiki Vishwavidyalaya, India - Bachelor Of Engineering, Computer Science

PROJECTS

Generative AI & RAG Toolkit for Web Summarization, Chatbots, and Knowledge Retrieval, [Link](#)

- Built **LLM-powered** tools for **webpage summarization**, **tech Q&A**, **meeting minutes generation**, and **knowledge retrieval** using **OpenAI**, **Ollama**, **ChromaDB**, and **FAISS**.

AI-Powered Virtual Fashion Try-On Platform,, [Link](#)

- Developed an AI-driven platform using **Stable Diffusion** and **DensePose** with **Model Quantization** to enable real-time virtual garment try-on, enhancing e-commerce experiences and empowering influencers with realistic fashion content creation.

SKILLS

Programming Languages	: Python, JavaScript, C++, C#
Web Development	: HTML/CSS, Flask, FastAPI, REST APIs, Django
Data Management	: SQL, MySQL, PostgreSQL, MongoDB, BigQuery
Cloud Technologies	: AWS (EC2, S3, RDS, VPC), GCP, Docker, Git, CI/CD (GitLab, Travis)
Data Engineering	: Apache Airflow, Apache Spark, Data Pipelines, ETL Automation, Pandas, Numpy, Matplotlib, Scrapy, BeautifulSoup
Machine Learning	: Scikit-learn, TensorFlow, Keras, NLTK, MLflow, LLMs, RAG
Tools & Platforms	: JIRA, Bitbucket, GitHub, Selenium, Linux, IBM Watson, Tableau